

Building AI-Driven Web Experiences at Scale

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Abstract

AI is no longer just a component of Web systems; it is increasingly shaping the experiences users have online. From search and recommendation to conversational and generative interfaces, AI is redefining how people interact with content at Web scale. In this keynote, I reflect on how recent advances in AI, including deep learning and generative models, are reshaping the design space of Web technologies and the systems that support them.

Drawing on insights from developing AI-driven systems at Spotify, I discuss how search and recommendation are evolving into interactive, intent-aware, and user-controllable experiences that support exploration and discovery. The talk highlights emerging system paradigms, deployment challenges, and open research questions around building such experiences at scale, and reflects on the implications for the design of future Web systems and interactions.

CCS Concepts

• **Information systems** → **Recommender systems**; **Web searching and information discovery**; • **Computing methodologies** → **Artificial intelligence**.

Keywords

Search and recommendation; Generative AI; Personalization at scale; Exploration and discovery

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BIOGRAPHY

Mounia Lalmas is Senior Director of Research at Spotify, where she leads Tech Research in Personalisation. Her work develops algorithmic and methodological solutions that help users connect with content they will enjoy across modalities such as music [5], podcasts [13] and audiobooks [12].

Her research spans information retrieval and search [6], recommender systems [11], evaluation [4, 16] and user modelling [1, 9, 15]. Lately her focus has been on the application of modern AI techniques, including generative models, to large-scale online systems [14, 17].

She previously held senior roles in industry and academia, including Director of Research at Yahoo, where she focused on advertising quality [2, 18], user engagement [3, 8], and search [10], and Professor of Information Retrieval at Queen Mary University of London and Research Professor at the University of Glasgow. Mounia is an Honorary Professor at University College London and a Distinguished Research Fellow at the University of Amsterdam.

An active member of the web, search, and recommender systems research communities, Mounia regularly serves on senior program committees for major conferences such as WSDM, KDD, and SIGIR. She has co-chaired ECDL 2010, SIGIR 2015, WWW 2018, WSDM 2020, and CIKM 2023, and has authored over 270 publications. She co-authored *Measuring User Engagement* [7], which emerged from a tutorial delivered at WWW 2013.

Her work bridges foundational research and real-world deployment, with a current focus on how AI is reshaping large-scale systems. She was nominated for the VentureBeat Women in AI Awards for Research in both 2022 and 2023 and is a member of the SIGIR Academy Class of 2024.



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